CARISURG MEDTECH PATHWAYS RESEARCH PROJECT

PRELIMINARY PROPOSAL

Project Title: An Explainable AI Assistant to Support Emergency Department Triage in the Caribbean

Student Name: Sariana Ramoutar

Statement of the Problem:
Emergency departments rely on rapid triage decisions to prioritise patient care, but busy clinical environments, resource constraints and human subjectivity can lead to inconsistent triage outcomes (De Freitas et al., 2020; Seo et al., 2024). In recent studies, artificial intelligence has demonstrated strong potential in improving triage speed and predictive performance. However, many systems function as "black boxes," meaning they provide recommendations without explaining the reasoning behind them (Da'Costa et al., 2025; Tyler et al., 2025). As a result, clinicians may find it difficult to understand, verify and confidently incorporate AI recommendations into routine decision-making, limiting their practical use in emergency care (Bhuvaneswari et al., 2025).

Previous Work:
Recent clinical literature highlights both the potential benefits and implementation challenges of AI within emergency triage.

- Multiple studies report that machine learning models frequently can outperform traditional triage approaches and support **patient prioritisation, admission prediction and resource planning** (Tyler et al., 2025; Da'Costa et al., 2025; Almulihi et al., 2024). Researchers have applied these models to predict immediate Emergency Severity Index (ESI) levels, downstream hospital admissions, ICU placements, and overall patient mortality risks (Agius et al., 2025; Almulihi et al., 2024; Tschoellitsch et al., 2023). Furthermore, advancements in Natural Language Processing (NLP) show that AI can **extract clinically relevant information from free-text nursing notes** that may be difficult for traditional systems to analyse (Lee et al., 2024).
- Triage accuracy is influenced by factors such as nurse experience and the amount of time available for patient assessment (Seo et al., 2024). Research suggests that clinicians often use "situational triage" to balance patient risks against operational pressures such as overcrowding and limited staffing. Observational research further demonstrates that emergency department workflows are shaped by staffing shortages, resource limitations and patient flow bottlenecks, requiring clinicians

to continually adapt their practices to maintain patient movement (De Freitas et al., 2020). These findings support the development of **human-centred AI frameworks** that complement existing workflows rather than disrupt them (Bhuvaneswari et al., 2025; Gorick et al., 2023).

- Despite promising performance in research settings, **several challenges remain before widespread clinical adoption**. First, some studies report a tendency toward over-triage, which may increase pressure on limited healthcare resources (Tyler et al., 2025). Second, **algorithmic bias remains a concern**, particularly when models are trained on datasets that do not adequately represent diverse patient populations (Da'Costa et al., 2025; Huang et al., 2022). Finally, many published systems have **limited external validation** and are trained on isolated, single-institution datasets, making it difficult to determine how well they would perform in different healthcare settings (Agius et al., 2025; Araouchi & Adda, 2024; Tschoellitsch et al., 2023). This limitation is reinforced by (Wong et al., 2021), who found that a widely deployed commercial AI system for detecting sepsis missed a large proportion of true cases while generating excessive false alarms when independently evaluated in a real hospital environment. Their findings highlight the importance of rigorous external validation before AI systems are introduced into clinical practice.
- Recent research also suggests that **technical performance alone is insufficient for successful adoption**. Clinicians are less likely to use AI systems that cannot explain how recommendations are generated, while poorly designed explanations may reduce trust or even encourage inappropriate over-reliance on AI recommendations (Amann et al., 2022; Rosenbacke et al., 2024). Furthermore, **safe deployment requires structured safety assurance** throughout AI development, including identifying potential hazards and incorporating safeguards before implementation in clinical environments (Festor et al., 2022). This limitation is particularly important for Caribbean healthcare settings, where workflow changes and resource availability may differ substantially from environments in which most AI systems have been developed and evaluated (De Freitas et al., 2020).

Collectively, these studies indicate that although strong predictive performance is achievable, successful implementation depends equally on explainability, clinician trust, external validation and patient safety. However, **a gap remains for transparent, explainable triage tools** that integrate seamlessly with existing clinical workflows while remaining appropriate for resource-constrained Caribbean emergency departments.

Gap Analysis:

Although recent studies demonstrate that AI can improve emergency department triage, two key gaps remain before these systems can be safely adopted in routine clinical practice.

First, many existing AI systems focus primarily on predictive accuracy while providing little explanation for how recommendations are generated (Amann et al., 2022; Tyler et al., 2025). Without understandable explanations, clinicians may struggle to determine whether an AI recommendation is reasonable, reducing trust in the system and limiting its practical use. Conversely, research also suggests that explanations must be carefully designed, as overly complex explanations can overwhelm users or encourage inappropriate confidence in incorrect recommendations (Rosenbacke et al., 2024).

Second, current research has largely compared AI performance against clinicians, rather than examining how AI can safely support clinicians in real emergency department environments. Successful implementation depends not only on prediction accuracy but also on supporting clinical judgement, integrating naturally into existing workflows and maintaining patient safety in busy, resource-constrained settings (Bhuvaneswari et al., 2025; De Freitas et al., 2020; Festor et al., 2022).

Together, these gaps highlight the need for AI systems that are both explainable and designed to support, rather than replace, clinical decision-making.

Proposed Solution:

This project will develop and evaluate an Explainable AI (XAI) decision-support prototype designed to assist emergency department clinician triage decisions using a de-identified, publicly available dataset. In contrast to traditional "black box" systems, the proposed model will simultaneously predict patient acuity levels and generate a visual explanation showing which clinical features and vital signs most influenced each recommendation. Clinicians will remain responsible for every final triage decision.

The proposed model aims to achieve triage classification performance comparable to that reported in previous benchmark studies (e.g., Araouchi & Adda, 2024), while introducing explicit visual explainability features to address the trust and transparency concerns highlighted in the literature (Amann et al., 2022; Da'Costa et al., 2025; Rosenbacke et al., 2024). This prototype will also incorporate key safety principles identified in recent AI safety research, ensuring that explainability supports informed clinical judgement without encouraging over-reliance on AI recommendations (Festor et al., 2022). The long-term goal is to develop a transparent, human-centred AI assistant that can be adapted to the workflows and resource constraints of Caribbean emergency departments.

Constraints & Stakeholders:

Successful implementation of an AI-assisted triage system depends not only on predictive performance but also on how well it fits within existing emergency department workflows while remaining safe and trustworthy for clinicians and patients. The following workflow constraints and stakeholder needs were identified in the literature and will guide the design of the proposed explainable AI prototype.

Workflow Constraints

1. Limited information available at triage

Emergency clinicians often make triage decisions before complete patient information is available. Laboratory results, previous medical records and additional clinical documentation may only become available later in the patient journey, creating downstream workflow bottlenecks (De Freitas et al., 2020). AI models must often rely on extracting meaning from highly variable, unstructured initial nursing notes instead of complete datasets (Lee et al., 2024). An AI-assisted triage system must therefore be capable of generating recommendations using only information typically available during the initial triage assessment, such as presenting complaints, vital signs and nurse observations.

2. High workload and time pressure

Emergency departments are fast-paced, highly resource-constrained environments where staffing shortages frequently occur, sometimes leaving shifts operating with less than half of their required nursing staff (De Freitas et al., 2020). Triage nurses must assess patients, document findings and manage patient flow within a limited timeframe, and triage accuracy is directly tied to the amount of evaluation time available (Seo et al., 2024). Any AI solution that requires additional data entry or complex interactions may increase workload and reduce adoption. The system should therefore integrate into existing workflows and provide recommendations with minimal additional effort.

3. Existing triage processes must be maintained

Emergency departments rely on established triage procedures to prioritise patient care and coordinate patient movement. Triage nurses frequently rely on holistic reasoning, clinical intuition and situational judgement rather than rigid algorithms (Gorick et al., 2023). Introducing additional approval steps or workflow interruptions could delay care and reduce efficiency. The proposed AI system should function as a decision-support tool within existing triage processes rather than replacing clinician judgement or changing current workflows (Amann et al., 2022; Bhuvaneswari et al., 2025).

Clinical Stakeholders

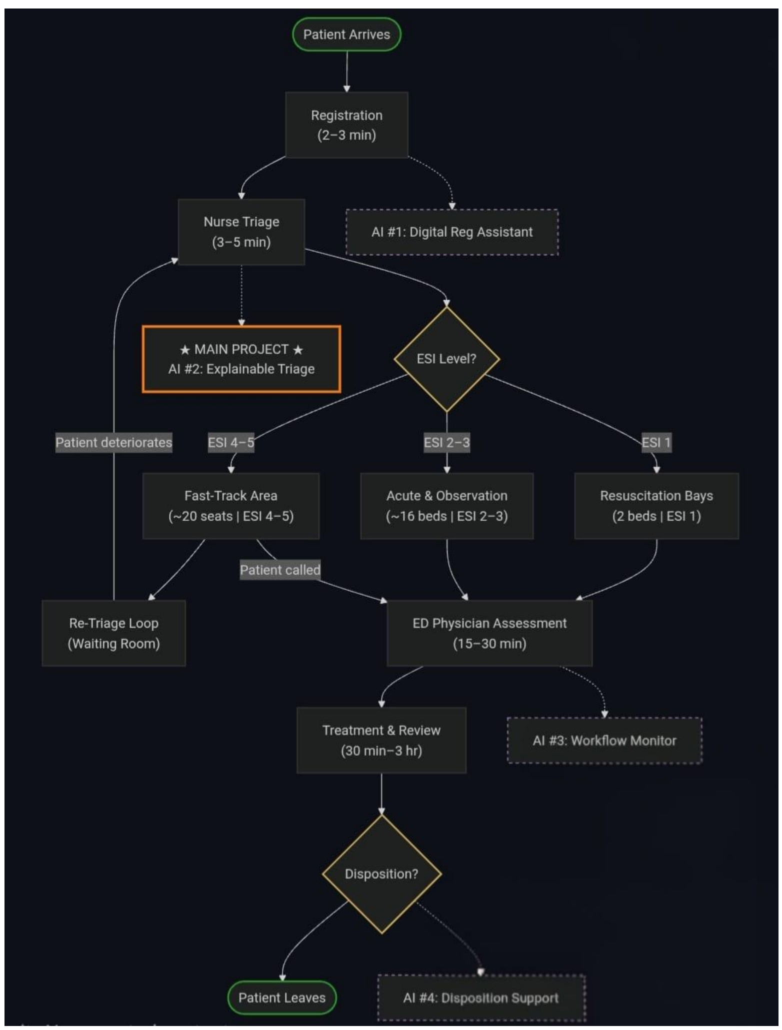
Stakeholder	Primary Concern	Why This Matters
Triage Nurse	Recommendations must be accurate, understandable and delivered quickly enough to support real-time decision-making without increasing workload.	AI recommendations must fit naturally into nurses' clinical reasoning and be delivered quickly enough to support time-sensitive decisions (Gorick et al., 2023; Seo et al., 2024).
Emergency Department Consultant	AI recommendations should improve patient prioritisation while maintaining clinician oversight and patient safety.	Clinicians remain legally and professionally responsible for patient care, so the AI should support – not replace – clinical judgement (Da'Costa et al., 2025; Bhuvaneswari et al., 2025).
Administrative Clerk	The system should integrate with existing registration and electronic health record processes without creating duplicate documentation.	Registration is the first stage of the emergency department workflow. The system should integrate with existing documentation processes rather than create duplicate work (De Freitas et al., 2020; Lee et al., 2024).
Porter / Support Staff	Faster identification of high-priority patients may improve patient flow and reduce unnecessary delays in moving patients between departments (De Freitas et al., 2020).	When porters are delayed, highly-trained clinicians are forced to "substitute down" to wheel patients to scans manually (De Freitas et al., 2020).
Patient / Patient Representative	AI-assisted triage should be transparent, fair and support timely access to care while ensuring clinicians remain responsible for final decisions.	Patients benefit these recommendations are reviewed by nurses before decisions affecting their care are made (Da'Costa et al., 2025; Rosenbacke et al., 2024).
Hospital IT / Digital Health Team	The system must integrate securely within existing hospital software while protecting patient data and maintaining reliable operation.	Safe integration, cybersecurity and system reliability are essential before any AI tool can be used in clinical practice (Festor et al., 2022).

<u>Mercer General Hospital Context</u>

Mercer General Hospital is a fictitious Caribbean regional referral hospital used as a representative case study for this project. It operates a nurse-led, five-level Emergency Severity Index (ESI) triage process in which patients are initially assessed using information available at presentation, including presenting complaint, vital signs and a brief clinical history, before being assigned to an appropriate treatment area. The proposed AI assistant is intended to support this initial triage stage while fitting within existing workflows and preserving clinician responsibility for all final triage decisions.

Workflow Diagram showing the Current Triage Process for Mercer ED:

This diagram maps the patient path through the Mercer General Hospital Emergency Department from arrival to discharge. It traces how patients are registered, assessed by a triage nurse, and assigned to a treatment area based on their ESI score. The diagram also marks four areas where AI can improve the workflow, with this project focusing specifically on **AI Opportunity #2 (Explainable Triage)** to assist nurses during the initial assessment.



Process Notes for the Workflow Diagram

- **Registration** (2–3 min): Demographic and administrative information is collected before clinical assessment begins. AI Opportunity #1 (Digital Registration Assistant) represents a possible future solution to streamline this stage but is outside the scope of this project.
- **Nurse Triage** (3–5 min): A triage nurse performs the initial assessment using information immediately available at presentation, including the chief complaint, vital signs and brief medical history. This stage determines the patient's ESI level.
- **AI Opportunity #2 (Main Project)**: This project focuses on providing explainable decision support during the initial triage assessment. The system is intended to assist nurses by explaining the factors influencing its recommendations while preserving clinician oversight.
- **ESI-Based Patient Routing**: Patients are directed to different treatment areas according to urgency. ESI 1 patients proceed directly to the resuscitation bays, ESI 2–3 patients are managed in acute and observation areas, and ESI 4–5 patients are placed in the fast-track area.
- **Re-Triage Loop**: Patients waiting in the fast-track area may deteriorate and require reassessment. Re-triage is part of standard clinical practice and allows patients to be upgraded to higher-acuity pathways when necessary.
- **Physician Assessment** (15–30 min) **and Treatment** (½–3 hr): Emergency physicians assume responsibility for diagnosis and management. Additional investigations, treatment and specialist consultations may occur before a disposition decision is made.
- **AI Opportunities #3 and #4**: Workflow monitoring and disposition support represent additional areas where AI could be applied. However, these functions occur later in the patient journey and are outside the scope of the proposed work.
- **Project Scope**: The proposed system operates only during the triage stage and does not replace clinical judgement, modify existing workflows or eliminate the need for re-triage and ongoing patient monitoring.

Meanings of Emergency Severity Index (ESI) Score

ESI	Meaning	Target Time to be Seen
1	Life-threatening – requires immediate intervention (cardiac arrest, major trauma, shock)	Immediate
2	High-risk, severe pain, disoriented, or vitals in danger zone (chest pain, suspected stroke)	≤ 10 min
3	Needs multiple resources to stabilise; vitals not in danger zone (sepsis, abdominal pain)	≤ 30 min
4	Needs one resource (simple laceration, urinary tract infection)	≤ 60 min
5	Needs no clinical resources (prescription refill, minor complaint)	Can wait

Risk Analysis:

Artificial Intelligence (AI) systems used in healthcare must be designed with patient safety, clinician oversight and ethical practice in mind. Although the proposed system is intended only as a clinical decision-support tool, there are still technical, operational, ethical and equity risks that could affect its safe implementation. Identifying these risks early allows appropriate safeguards to be incorporated into both the system design and evaluation process. The table below summarises the principal risks identified for the proposed Explainable AI (XAI) triage assistant, together with their likelihood, potential impact and planned mitigation strategies.

Risk Register

Risk [Category]	Likelihood	Impact	Mitigation [Success Indicator]
1. Incorrect Triage Recommendation [AI-Technical]	Medium	High	Validate the model before each testing phase using a separate dataset. The AI provides recommendations only, while the triage nurse remains responsible for the final ESI decision. The interface explains the clinical factors influencing each recommendation. [Validation testing shows fewer than 2% of high-acuity (ESI 1–2) patients are under-triaged, and nurses override incorrect recs when appropriate.]
2. Poor Performance on Caribbean Patients [Equity]	Medium	High	Evaluate model performance across available patient groups and clearly document any limitations before deployment. Future versions should be validated using Caribbean clinical data. [Performance differences between patient groups remain small, or any limitations are clearly documented for future improvement.]
3. Misleading AI Explanations [AI-Technical]	Low	High	Present explanations using clear, non-technical language that highlights the patient's most influential symptoms and vital signs. Review explanations during testing to ensure they accurately reflect the model's reasoning. [Clinicians report that explanations are easy to understand and accurately represent the AI's reasoning during user testing.]

4. Increased Triage Time [Operational]	Medium	Medium	Use information already collected during routine triage to avoid duplicate data entry. Measure assessment time during usability testing and simplify the interface if delays occur. [Average triage time increases by less than one minute compared with the existing workflow.]
5. Over-Reliance on AI Recommendations [Ethical]	Medium	High	Clearly state that the AI is a decision-support tool only. Nurses remain responsible for every triage decision and are encouraged to compare the AI's recommendation with their own clinical judgement. [Clinicians continue overriding AI recommendations when appropriate, demonstrating that independent clinical judgement is maintained.]
6. Patient Privacy and Data Security [Ethical]	Low	High	Use only de-identified, publicly available datasets during development. Restrict access to project data and exclude all patient-identifying information from the prototype. [No patient-identifiable information is used or stored during development or testing.]
7. Unfair Recs for Certain Patient Groups [Equity]	Medium	High	Compare prediction accuracy across different patient groups during evaluation and investigate any significant differences before further development. [No patient group performs substantially worse than the overall model, or any differences are clearly reported.]
8. Clinicians Do Not Trust the AI [Operational]	Medium	Medium	Focus the prototype on explainability by showing why each recommendation was made. Collect clinician feedback after testing and improve explanations where necessary. [Most clinicians report that they understand the AI's recommendations and find the explanations useful.]
9. AI Fails to Work at Other Hospitals	Medium	High	Clearly state that the prototype has only been validated on the available dataset. Future deployment should

[AI-Technical]			include validation using local hospital data before clinical use. [Project documentation clearly identifies external validation as a requirement before deployment in another hospital.]
10. System Failure or Technical Downtime [Operational]	Low	High	Ensure existing manual triage procedures remain available at all times so patient care can continue if the AI becomes unavailable. [During testing, manual triage continues safely whenever the AI system is unavailable.]

Key Risks

1. Incorrect Triage Recommendation

The greatest clinical risk is that the AI may recommend an incorrect ESI level, potentially delaying treatment for critically ill patients or unnecessarily prioritising lower-acuity cases. To reduce this risk, the proposed system functions only as a decision-support tool, with the triage nurse remaining responsible for the final decision. Explainable AI techniques are also incorporated so that the clinical factors influencing each recommendation are visible to the user, allowing inappropriate recommendations to be identified before clinical action is taken (Wong et al., 2021; Festor et al., 2022).

2. Clinicians Do Not Trust the AI

A second important risk is poor clinician acceptance. Healthcare professionals are unlikely to adopt an AI system if its recommendations cannot be understood or justified. This project addresses that challenge by prioritising explainability rather than prediction accuracy alone. Each recommendation is accompanied by an explanation of the symptoms and vital signs that contributed most strongly to the prediction, enabling clinicians to judge whether the recommendation is reasonable and improving confidence in the system (Amann et al., 2022; Rosenbacke et al., 2024).

3. Over-Reliance on AI Recommendations

Although explainability can improve trust, excessive reliance on AI introduces a separate risk known as automation bias. Clinicians may gradually begin accepting recommendations without sufficient independent assessment. To minimise this risk, the proposed system is designed to augment rather than replace clinical judgement. Nurses retain full responsibility for every triage decision, and the AI serves only as an additional source of information to support clinical decision-making rather than acting as the final authority (Rosenbacke et al., 2024).

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June 30th, 2026.